

Matrix Computation Homework 4

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Problem 1. (Quadratic Form)

1)

Solution.

It is obviously that:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & a \end{bmatrix}$$

2)

Solution.

First, we find the eigenvalues by solving the characteristic equation $|\mathbf{A} - \lambda \mathbf{I}| = 0$:

$$\det \begin{bmatrix} 2 - \lambda & 1 & 1 \\ 1 & 2 - \lambda & 1 \\ 1 & 1 & a - \lambda \end{bmatrix} = 0$$

It is obviously that:

$$(1 - \lambda)(\lambda^2 - (3 + a)\lambda + 3a - 2) = 0$$

Thus we can get that $\lambda_1 = 1$ and $\lambda_2 + \lambda_3 = 3a - 2$, $\lambda_2 \cdot \lambda_3 = (3 + a)/2$

Since $\mathbf{A}$ is PSD, we need to ensure:

$$\lambda_2 + \lambda_3 \geq 0 \quad \text{and} \quad \lambda_2 \cdot \lambda_3 \geq 0$$

Thus we can get:

$$a \geq \frac{2}{3}$$

3)